

SURVEY

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A review of distinct machine learning classifiers for healthcare fraud detection

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Abstract

Healthcare insurance fraud is a major problem, with an estimated \$300 billion lost annually in the United States. Machine learning has been explored as a tool for fraud detection for over a decade, but challenges remain, including class imbalance, dataset diversity, and model interpretability. This survey reviews 22 supervised, unsupervised, and semi-supervised classification techniques published between December 2017 and October 2024 which are novel within the healthcare fraud detection domain. Supervised techniques, which make up a majority of the identified works, are divided into deep learning, graph-based, and meta-learning methods. We also find a small set of novel unsupervised and semi-supervised classifiers, which we examine with a focus on model explainability. We find that deep learning is a popular and effective approach for supervised learning, while research gaps remain in areas such as transfer learning and incremental learning for all three classification approaches. Most concerningly, we find few to no works which establish benchmarks across diverse techniques. We assert that addressing these gaps could lead to more effective, transparent, and adaptable fraud detection systems in healthcare insurance.

Keywords: Survey, Literature review, Fraud detection, Healthcare fraud detection, Supervised learning, Unsupervised learning, Semisupervised learning, Deep learning

Introduction

In the domain of health insurance, fraud is a significant issue. It is estimated that as much as \$300 billion is lost every year to health insurance fraud in the United States alone [1]. Due to the high volume and velocity of health insurance claims, only a small amount of fraud can be manually investigated; in 2020, only \$3.1 billion of false and fraudulent claims were recovered [2]. However, machine learning methods can be used to create detection systems which can automatically (and potentially more accurately) find fraud, which could help prevent unnecessary losses. Given the potential value of effective automated fraud detection systems, machine learning techniques for fraud detection have been studied as early as the 1980s [3].

However, there are many challenges with developing machine learning techniques in this domain. For one, healthcare fraud datasets can be very large (often encompassing millions of claims), which necessitates efficient classifiers. Additionally, severe class imbalance (i.e., the rarity of fraudulent claims compared to non-fraudulent ones) in most

of these datasets can inhibit the accuracy of many models without careful preprocessing. Complexity also arises from the diversity of datasets, with many having complex, non-numerical features difficult for machines to interpret. Finally, the “explainability” of a machine learning model’s predictions are important when the model is applied in the real world with high stakes (as would be the case for large-scale fraud detection models); however, many complex models, such as deep neural networks, make predictions in a “black-box” fashion which is difficult or impossible for humans to interpret.

Machine learning classifiers—that is, techniques which categorize data into one of multiple classes, such as fraudulent or non-fraudulent—can be separated into three main types. Supervised classifiers learn from “labeled” data (that is, data which is known to be fraudulent or benign) in order to categorize future unlabeled data. Unsupervised classifiers operate with only unlabeled data (that is, claims which may or may not be fraudulent), learning to identify patterns from other dataset features. Finally, semi-supervised classifiers (also called hybrid models) combine both approaches, training from a mix of labeled and unlabeled data; most often, an unsupervised method (not always a classifier) is used to identify patterns which are then used to help train a supervised model. Generally speaking, supervised and unsupervised techniques are the most common in machine learning literature, while semi-supervised techniques are more rare.

A recent survey paper by Leevy et al. [4] reviews the latest unsupervised techniques for anomaly detection for health insurance fraud. Our work is an extension of that review; however, rather than focusing on unsupervised learning, we conduct a review of machine learning within the health insurance fraud detection domain as a whole. To our knowledge, the most recent comparable review of machine learning approaches for healthcare fraud was a survey by Bauder et al. from 2016 [5].

In 2019, Ikono et al. [6] provided a meta-analysis of fraud detection in healthcare; however, they only report trends and observations about relevant techniques and do not detail specific algorithms or approaches used. Similarly, Du Preez et al. [7] provide a systematic literature review of machine learning approaches for fraud detection in healthcare claims, also covering supervised, unsupervised, and semisupervised learning. However, their review broadly covers all relevant works over a long timespan, and focuses largely on statistical analyses (e.g., the frequency of different categories of approaches over different time periods). Several comparative studies have also been done since 2016, but were not intended as surveys and thus were limited in scope.

In contrast to these works, our goal is to highlight the wide variety of techniques which have been applied in this domain; thus, we place an emphasis on recent and novel methods, and go into significantly more depth on the methodology and results of each relevant work. Other than Leevy et al.’s review of unsupervised techniques [4], we find no recent literature conducting a survey of this type in the application of healthcare fraud detection.

The objective of this review is to list and discuss novel and domain-novel classifiers specifically used in healthcare fraud detection in the recent past (from December 2017 to October 2024). A “classifier” is defined as a machine learning model which categorizes data into one of multiple classes; other machine learning techniques used for data pre-processing or analysis, such as data sampling or clustering, are not relevant to our review. We use the term “domain-novel” to describe any technique which was

initially utilized in another domain, but was applied experimentally for the first time in this domain. The distinction between “domain-novel” and “novel” helps clarify whether a given study has invented and proposed a completely new technique, or if it has simply introduced an existing technique to the healthcare fraud detection domain (both of which are relevant to our review).

The remainder of this paper is organized as follows. “Methodology” provides details on the process by which studies were selected for review. “Results” describes all studies found in our review, with subsections corresponding to different categories of techniques (e.g., deep learning, or hyperparameter optimization). “Discussion” discusses and analyzes the works found in each category, and their respective shortcomings. Lastly, “Conclusion” recaps our findings, and concludes with recommendations for future work.

Methodology

Google Scholar was used as the primary source for the literature covered in this review. We performed multiple searches to find all recent studies (between December 2017 and the time of search, October 2024) introducing novel or domain-novel classification techniques in the field of healthcare fraud detection. A few examples of Google Scholar queries used in this study are listed below:

- medicare fraud detection
- healthcare fraud anomaly detection machine learning
- “supervised machine learning” AND (intitle:“healthcare fraud” OR intitle:“health fraud” OR intitle:“medical fraud” OR intitle:“insurance fraud”)
- “supervised learning” AND (intitle:“fraud” OR intitle:“anomaly detection”) AND (intitle:“medicare” OR intitle:“healthcare” OR intitle:“health insurance” OR intitle:“medical insurance”) AND (“novel” OR “new” OR “innovative” OR “proposed method”) -intitle:“automobile” -intitle:“auto insurance” -intitle:“unsupervised”

The full results of these searches were then reviewed in detail, and a subset of relevant studies was then selected. A study was considered to be relevant if its primary focus was to introduce a novel (or domain-novel) classifier in the field of healthcare fraud detection. We excluded from our review any studies which describe machine learning techniques which have previously been applied to healthcare fraud detection; any studies which propose machine learning techniques which are not classifiers; and any studies which propose machine learning techniques for applications other than healthcare fraud detection. Any papers already reviewed and discussed by Leevy et al.’s survey on unsupervised techniques [4] were also excluded. In total, this search and exclusion process yielded 22 relevant papers, which we describe and analyze throughout the rest of this work.

Results

When discussing baseline models used as comparison points for each work's proposed model, we use the term State-of-the-Art (SOTA) to refer to those baselines which give strong performance for the relevant application, are relatively recent or contemporary, and are roughly comparable to the proposed model in question. We distinguish these baselines from those which are exceedingly simple or outdated (such as a simple linear regression model), as non-SOTA baselines often cannot provide a full perspective on the performance of a given model compared to other available options.

Supervised learning

Supervised classifiers are those which train from a dataset which is fully labeled; i.e., the class of each instance is known and provided to the model during training.

Although the extra information provided by labels is useful for learning, many obstacles arise in fully utilizing this information. The most prominent issue is that fully labeled datasets are difficult to acquire, as real data (rather than synthetically created data) must be labeled using the knowledge (and often time) of human experts. Further, the labels within these datasets are inevitably somewhat inaccurate, especially for applications such as fraud detection, where only a small fraction of real-world fraud is actually caught. In most cases, a portion of fraudulent claims will be incorrectly labeled as non-fraudulent, and vice versa; this is commonly known as “class noise” [8]. Additionally, healthcare datasets often feature severe class imbalance (where one minority class is vastly outnumbered by one or more majority classes)—without proper countermeasures, imbalanced training data will often lead to models biased towards the larger majority class [9]. Supervised classifiers in this domain must be designed to handle these types of issues.

In this section, we discuss the relevant supervised classifiers found in our literature search. We divide this subsection into smaller categories based on the nature of each approach.

Deep learning approaches

Deep learning is a category of machine learning approach which utilizes many layers of artificial neural networks, or neural network variants such as Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs). The use of many layers often allow deep learning models to construct new representations of input data which are more useful for classification but significantly less interpretable by humans. Certain algorithmic innovations, such as the attention mechanism [10] and Long Short-Term Memory (LSTM) [11], have dramatically improved the performance of deep learning models. These new techniques are often able to be used modularly, allowing for a wide range of possible neural network architectures. Additionally, improved hardware has drastically accelerated training time for large models. Some such architectures have been utilized for healthcare fraud detection and are described below.

Ding et al. [12]

This study proposes the Credit Evaluation Model (CEM) for detecting health insurance fraud. The approach involves constructing a novel data structure known as the

Credit Memories Module (CMM), which contains a list of credit vectors corresponding to healthcare institutions. By dynamically updating the CMM using LSTM and Linear Discriminant Analysis (LDA), the proposed technique is better able to utilize the history of an individual's healthcare claims to predict their fraudulence. The researchers apply a base classifier, such as Random Forest (RF), to tentatively classify instances as normal, fraudulent, or edge cases. The CEM then reclassifies the detected edge cases and updates the CMM with fraudulent cases (improving the model's efficiency later). The researchers used a labeled dataset containing five million healthcare insurance claim instances from 40,000 patients, proposing a novel algorithm which determines claim-level labels from the given patient-level labels, and generates synthetic fraud labels if not enough are found. To test the effectiveness of the proposed CEM, the researchers augmented three baseline decision tree models (RF, XGBoost (XGB) [13], and LightGBM [14]) with their technique and observed changes in performance between the base and modified models. The CEM techniques significantly improved each base model in F1 score and Area Under the (Receiver Operator Characteristic) Curve (AUC) score, and CEM combined with LightGBM (0.90 F1) outperformed another LightGBM variant named LHEM [15] (0.85 F1).

Farbmacher et al. [16]

This study aimed to create an explainable attention network for healthcare fraud detection. They used a dataset from a private health insurer in Germany with nearly 400,000 claim instances, where 1.5% are labeled as fraud. The authors highlight the various challenges presented by this data, namely categorical features with high cardinality, a variable input size (due to variable claim length), and a significant class imbalance. The authors propose a deep learning feature extraction process which addresses each of these problems using categorical embeddings, an aggregation layer with attention mechanisms, and cost-sensitive learning, respectively; a neural network is then used to classify the extracted features. The proposed method was tested against two other methods: random guessing (their "weak benchmark") and boosted trees (BT). The researchers showed that their proposed deep learning model outperformed these baselines in terms of proportion of "top" fraud claims detected, as well as in terms of hypothetical savings from model predictions. The researchers argue that their model is explainable because the attention weights could potentially elucidate what items in a claim are relevant to the model's prediction.

Matschak et al. [17]

The authors of this study present a novel CNN-based approach for healthcare fraud detection. The researchers claimed that this was the first application of CNNs in the context of medical claim fraud detection, since this model type is normally used for image processing tasks. The proposed technique converts vectorized claim instances (grouped by provider) into a 78x78 grid format, then utilizes convolutional and dense neural network layers to predict seven distinct categories of fraud. The technique was also designed to handle multi-class classification, which allows for distinguishing between different types of fraud such as patient abuse, license revocation, and others. Experiments were run on a provider-level dataset from Medicare Part B (labeled using the List of Excluded Individuals and Entities (LEIE)), and a combination of oversampling and undersampling was used to balance the dataset from its original ratio of 10,000:1 to

a balanced ratio of 1:1. No baseline models were considered for comparison; multi-class predictions were converted to binary predictions for performance evaluation using a “one-vs.-all” approach. The researchers determined that the proposed CNN-based technique performed reasonably well (0.70 AUC) for the largest fraud type, while for other fraud types the model performed poorly (0.43-0.62 AUC); however, these judgements are subjective without a concrete point of comparison.

Mayaki and Riveill [18]

This study proposes a deep multi-input neural network model called MINN-AE. The proposed architecture uses two “blocks” in parallel to separately but simultaneously process different input types: a neural network block for item-level data (claims), and an autoencoder block for group-level data (providers). Then, the outputs of these layers are concatenated and fed to a final neural network block which makes final predictions on fraud. The novelty of this architecture is that it uses multiple data sources in parallel, without mixing them together in a single dense network. Experiments were run on six datasets: the Medicare Part B and Part D datasets (again labeled using the LEIE), the Healthcare Providers fraud detection dataset from Kaggle’s repository [19], and three other datasets from other fraud detection domains; all except the Kaggle dataset (with a 9% minority class proportion) had severe class imbalance (<1%). Ten-fold cross validation was performed, AUC and Area Under the Precision-Recall Curve (AUPRC) were used as performance metrics, and four traditional baseline models were built. Experiments also tested variants of each model with Random Undersampling (RUS) or alternative loss functions, to assist with the class imbalance problem. Direct comparison yielded mixed results, with a simple baseline neural network model often outperforming MINN-AE for given datasets, metrics, and augmentations; however, MINN-AE achieved the overall highest AUPRC score found for both healthcare datasets.

Graph-based approaches

A significant portion of the supervised techniques use knowledge graphs to represent and process data. Graph-based techniques oftentimes (but not always) exhibit high computational expenses in return for high performance.

From the papers, we found that the studies regarding these approaches usually start by constructing a knowledge graph from a dataset. Knowledge graphs represent information as connections between nodes. These graphs may be homogeneous, with only one type of node; for example, nodes on the graph could represent claim instances, and connections (edges) between nodes could represent claims sharing common features, such as provider type or procedure code. More commonly, graphs are heterogeneous, with different types of nodes; for example, a node could represent a patient, a hospital, or a date, and edges on the graph could represent connections between these entities. Once a graph is constructed from a dataset, it can then be used as a more efficient knowledge representation to train a model.

Chen et al. [20]

In this study, the authors propose a spatiotemporal constraint graph neural network (StGNN) technique for medical insurance fraud detection. The proposed method focuses on utilizing multi-semantic meta-paths (pathways through the graph that connect two patients, such as “patient-date-patient” or “patient-date-medicine-date-patient”) to

capture “behavioral trajectories” hidden in a heterogeneous medical insurance graph, then aggregate neighboring nodes with those trajectories via an attention mechanism. The method also constrains these trajectories both spatially, by prioritizing nodes which are more closely connected in the graph using an attention mechanism; and temporally, by dividing the graph by time period and forming each sub-graph into time series data. The study uses a real-world dataset, nicknamed “Medical-2,” from an unnamed insurance bureau involving more than 3 million detailed records of over 10,000 patients and moderate class imbalance (class ratio of 1:70) – a version of this dataset undersampled to a 1:2 ratio, nicknamed “Medical-1,” was also utilized. Compared to six other benchmark methods, including Logistic Regression (LR), RF, and four similar SOTA baselines, StGNN demonstrates superior performance across several metrics and dataset perturbations in both node classification and anomaly detection tasks .

Lu et al. [21]

Lu et al. author two papers which propose novel graph-based supervised classifiers for health insurance fraud detection; the first of these proposes a model based on a multi-level attention mechanism, named MHAMFD. This technique aims to utilize behavioral relationships mined from an attributed heterogeneous information network (AHIN) to construct a multi-relational graph network, which is then passed through a multi-level graph neural network to detect fraud. The researchers tested their model on the Medical-1 and Medical-2 datasets also used by Chen et al. [20], with respective class ratios of 1:2 and 1:70 (but notably did not compare their model to the StGNN model proposed in [20]). Additionally, several splits of data were made from different ratios for training, validation, and testing. The proposed technique was tested against four other graph-based healthcare fraud detection techniques over various performance metrics. MHAMFD outperformed the other models in terms of F1 score and accuracy, regardless of the data imbalance or splitting ratio; ablation studies also confirmed that the model’s hierarchical attention component significantly improved its performance.

Lu et al. [22]

The second relevant paper by Lu et al. proposes a multibehavioral pattern-based graph neural network (MBGNN) to detect healthcare fraud, which the authors claim to have high performance, efficiency, and explainability. First, a heterogeneous graph relating patients, medicines, doctors, and hospital departments is decomposed into a set of behavioral patterns using a technique the researchers named “multibehavioral pattern decomposition.” Then, a “multipattern fusion” component aggregates intrapattern, interpattern, and topological information into a fixed-size vector. Finally, a multilayer perceptron is used to predict fraud, and cross-entropy loss is used. The system is intended to be interpretable due to its use of an attention mechanism, which indicates how important certain patterns are towards making a prediction. The researchers again utilized the Medical-1 and Medical-2 datasets, and compared MBGNN to five similar baseline methods (including StGNN [20] but not MHAMFD [21]) using F1 score and precision. The proposed technique consistently outperformed the baseline models in all scenarios.

Hong et al. [23]

In this study, the researchers used a multi-channel heterogeneous graph structure learning (MHGSL) approach to detect health insurance fraud. First, a feature graph is made using K Nearest Neighbors (KNN) and cosine similarity between node features.

Then, a heterogeneous graph is constructed from the health insurance data, connecting patients, times, medical departments, and medicines, which is made into a semantic graph using meta-paths. Additionally, in order to preserve relationships between nodes and structural information, a topology graph is constructed using topological relations between nodes in the heterogeneous graph. Once all the graphs are constructed, they each go through both a specific graph convolution and shared parameter graph convolution. This creates six outputs from the convolution layer (three from the specific graph portion, three from the shared parameter graph portion), which are aggregated by an attention layer, and then reduced with a linear layer to reduce the dimensionality to the size of the node category. Finally, the output of this linear layer is normalized with the softmax function to get an encoded prediction. To test their model, the researchers test the proposed technique on the Medical-1 and Medical-2 datasets, and compared F1 score and accuracy with several other graph-based algorithms, including StGNN [20] and MHAMFD [21]. Overall, the proposed MHGSL method outperformed the other deep learning models in terms of F1 score, AUC, and accuracy (though StGNN and MHAMFD showed competitive accuracy in specific scenarios).

We note that the four previous graph-based classifiers were evaluated for the exact same task and dataset (anomaly detection on the imbalanced Medical-2 dataset), making their results uniquely comparable to one another. StGNN [20] achieves 0.875 F1, MHAMFD [21] achieves 0.881 F1, MBGNN [22] achieves 0.891 F1, and finally MHGSL [23] achieves 0.895 F1. Each successive paper yields a marginally higher performance than the last, indicating gradual improvement over time. However, none of these works use statistical tests such as Analysis of Variance (ANOVA) to confirm whether these improvements are statistically significant.

Chen et al. [24]

In this work, the authors develop a framework for automatic medical fraud detection (AMFD) for deployment in the healthcare industry. The researchers proposed a novel method named “Variational AutoEncoder-based Relational Model” (VAERM). This technique addresses the issue of limited fraud labels by taking advantage of an anonymized patient-doctor relational network, to supplement the one-class labels (taken from only the fraudulent data) used as the primary training data. This approach utilizes a graph convolutional neural network to feed the doctor-patient relational network into a deep neural network. Then, the model utilizes a Variational Auto-Encoder (VAE) [25] to predict a node in the tree as fraudulent or not. This study utilizes a medical fraud dataset collected from Shenzhen Nanshan People’s Hospital in 2015, filtered to a subset of instances describing one thousand doctors and 34 thousand patients: only 138 fraudulent patients were included, demonstrating extreme data imbalance. The study compares the proposed technique with 10 other deep learning techniques, some of which were also able to take advantage of the patient-doctor network, such as GraphSAGE [26] and a combination model created by the authors named GraphSAGE-VAE. The VAERM technique significantly outperformed all other models in terms of precision, F1, and accuracy, outperforming the best baseline, GraphSAGE-VAE, by a wide margin (0.928 F1 vs. 0.843 F1).

Mardani and Moradi [27]

In this study, the authors propose a novel technique for healthcare fraud detection which utilizes Graph Attention Networks (GATs) [28] for embedding relationships. The aim of this technique is to capture the complex relationships with a graph-based model while avoiding the high computational costs incurred by many graph-based techniques. A GAT works by assigning attention weights to neighboring nodes in a graph; in this study, graph edges representing either physicians and beneficiaries are processed separately, using two parallel GAT embedding blocks and two neural network blocks. Lastly, a final feed-forward neural network is used to combine and convert the resulting embeddings into a prediction, for which loss is calculated using cross-entropy. The researchers used Kaggle's Healthcare Provider Fraud Detection Analysis dataset [19]. To evaluate the performance of the proposed technique, the researchers compared their model to their implementation of a graph transformer network (GTN) [29] and some conventional algorithms – RF [30], XGBoost [13], and LR – using precision, recall, and F1 score. The proposed GAT-based model and the GTN model were also evaluated with many different hyperparameter settings to find optimal values. The researchers found that the GTN model was able to marginally outperform all baselines, achieving a 0.53 F1 score as compared to RF's score of 0.52. The authors also claimed that the GTN model exhibited extreme overfitting, while the GAT-based model showed better generalization; however, the authors did not provide any training-data results to demonstrate this claim.

Yoo et al. [31]

This study adapts an existing algorithm, GraphSAGE [26], for use in a Medicare fraud detection model. GraphSAGE (Graph Sample and Aggregate) is an inductive graph embedding framework, able to create embeddings for previously unseen nodes or sub-graphs not in the original training data. The researchers utilized Kaggle's Healthcare Provider Fraud [19] dataset, which also contains data on patient-provider and provider-provider relationships, and is only mildly imbalanced (class ratio of $\sim 1:9$). The model functions by modeling the tabularized patient-provider relational data as a graph, then using GraphSAGE to classify nodes in a graph as fraudulent or normal. The researchers evaluated the proposed model against a simple LR baseline in terms of four performance metrics: precision, recall, F1-score, and AUC. The proposed model achieved an F1 score of 0.49, significantly outperformed the baseline in all metrics except precision (where it only had an advantage of 0.01 points).

Zhang et al. [32]

In this study, the researchers present a pre-trained online contrastive learning (POCL) model for healthcare insurance fraud detection. Online learning describes when a model continuously learns from new, incoming data; the researchers claim this is the first work to introduce an online learning model for insurance fraud detection. First, a positive and a negative graph are constructed from the dataset, which are then used to build a pre-trained GraphSAGE model [26]. The output of the pre-trained model is then fed into a (non-pre-trained) GAT model [28] for final predictions, with a modified cross-entropy loss function to mitigate catastrophic forgetting during fine-tuning. To detect new fraud patterns as they emerge, a novel method called Temporal Memory Aware Synapses (MAS), which adaptively “forgets” less relevant fraud patterns and learns new ones, was incorporated into the model. The researchers utilized an enriched version of the Kaggle Healthcare Provider Fraud dataset, manually labeled to indicate whether each case of

fraud is at an individual or organized scale [33]. At a claim level, this dataset shows little to no class imbalance (38.1%). POCL was compared to five graph-based models and three variants of itself in terms of AUC, F1 score, accuracy, and time and memory consumption; each was tested over several months of simulated time.

The proposed technique achieved an F1 score of 0.635, showing significantly better performance, plus comparable time and memory efficiency, to the SOTA models. RetrainGNN, a variant of the proposed model which is retrained each time new data is introduced, performed marginally better than the base POCL (0.638 F1), but incurred severe time and memory costs. The proposed techniques were also found to be robust over time, demonstrating only mild dips in performance over the period of several simulated months.

Meta algorithms

A meta algorithm improves the performance of machine learning models by modifying how they process data or adjust their parameters. Typically, meta algorithms will “wrap” an existing algorithm, then modify its inputs, outputs, or weight updating rule to enhance performance. For example, “bagging” is a simple meta-algorithm that trains several copies of a learner on different subsets of a dataset, then combines their outputs to make a more accurate prediction than any one learner alone.

Hancock and Khoshgoftaar [34]

In this study, the researchers evaluated the CatBoost learner [35] on the task of Medicare fraud detection. CatBoost is part of a family of learners called Gradient Boosted Decision Tree (GBDT) algorithms, which create an ensemble of tree learners and use a specialized boosting algorithm to optimize the aggregated prediction; CatBoost is distinct from other GBDTs in that it can effectively process non-numeric features such as procedure codes. CatBoost also uses balanced decision trees and ordered boosting, which both alleviate model overfitting. The common, highly imbalanced Medicare Part B dataset (with the LEIE used for labeling) was used. Another GBDT algorithm, XGBoost, was used as a baseline model for comparison with AUC as the metric. Since XGBoost cannot process categorical variables on its own, it was evaluated on a modified version of the dataset aggregated to the provider-level with 3.7 million instances and 126 features; CatBoost was evaluated on the original claim-level version with 37 million instances and 8 features. RUS was applied to balancing both datasets from a >10,000:1 ratio to a completely balanced 1:1 ratio, and stratified 5-fold cross-validation was used to quantify the effect of randomness on results. CatBoost achieved a mean AUC somewhat greater than XGBoost (0.785 vs. 0.762 AUC); additionally including a categorical feature representing the provider’s home state boosted performance significantly (to 0.890 AUC). Analysis of performance standard deviation (<0.012 for all models) indicated that these differences were statistically significant and not the result of random chance.

Shi et al. [36]

In this paper, the authors experiment with functional principal component analysis (FPCA) and cost-sensitive learning (domain-novel techniques) for constructing a medical insurance fraud detection model, which are novel to the domain of medical insurance fraud. The proposed framework aggregates claim data into “temporal trajectories,” uses FPCA to convert these trajectories into a sequence of scores, and uses these as training

input for a classifier. The classifier is then wrapped in a cost-sensitive learner, which assigns greater importance to false negative results based on the cost of the prescribed drug, so as to minimize monetary losses from fraud. The researchers used multiple datasets, including Medicare Part D, which has very severe data imbalance (0.025% of instances are fraudulent). RUS was used to create five sampled datasets with varying levels of imbalance (between 1% and 50% minority class). To test the proposed framework, the researchers evaluated four classifier sub-models – Logistic Regression (LR), Random Forest (RF), a gradient boosting machine, and a neural network – both with and without cost-sensitivity, over F1 score and AUC. The performances of these models were also expressed as monetary costs, to demonstrate the actual savings the cost-sensitive learners would facilitate. Overall, the researchers found that the cost-sensitive learners were more capable of minimizing monetary losses than the non-sensitive learners, and that of the cost-sensitive learners, RF had the best cost-saving performance on average.

Yao et al. [37]

The authors propose an improvement upon the bagging algorithm called WTBagging, which incorporates a weighted threshold method. Bagging is a type of ensemble model which trains several models on smaller “bags” of a large dataset. In the standard bagging procedure, each model makes a classification for each instance (in this case, either fraudulent or non-fraudulent), and the majority vote is taken from the set of predictions as the final output. In the WTBagging algorithm, each bagged model is assigned a weight based upon its F1 score on a training set. If the sum of the weighted predictions of each model is greater than a certain threshold, the WTBagging model predicts fraud, and not fraud otherwise. To test their model, the researchers utilized the Medicare provider dataset, with labels attached using the LEIE. RUS was used to make three more evenly balanced datasets with ratios 90:10, 60:40, and 50:50; ten runs of five-fold cross-validation were also applied. Four “base” model types were used (RF, GBDT, XGBoost, and CatBoost), and five different configurations of both WTBagging and standard bagging were constructed using combinations of these base models. The results indicated that on average, the WTBagging models (which reached a maximum F1 score of 0.762) always performed better than the default bagging models (maximum F1 of 0.756), which in turn were always better than the GBDT baseline (0.740 F1); ANOVA was used to confirm the high statistical significance of these results.

Hyperparameter optimizers A hyperparameter optimizer is a type of meta-learner which iteratively updates a base learner’s hyperparameters (i.e., model settings) to find optimal values for a given dataset. There are a variety of strategies for optimizing (or “tuning”) hyperparameters, including random searching [38] and grid search (exhaustive search) [39].

We note that while many other papers change their model hyperparameters from their default settings in order to improve performance, this is distinct from hyperparameter optimization as its own meta-learning technique.

Arockiam and Pushpanathan [40]

The authors of this study propose a novel machine learning technique, which they call a MapReduce-based iterative support vector machine (MR-ISVM), for use in healthcare insurance fraud detection. The proposed model uses MapReduce

to preprocess and aggregate data to improve efficiency, then iteratively trains Support Vector Machine (SVM) models until a stopping condition (based on the Tanimoto index, a.k.a. the Jaccard index [41]) is met. The researchers utilized Kaggle's Healthcare Provider Fraud Detection Analysis dataset [19] to train their model, as well as two baseline SVM models with linear or radial basis function kernels. Each learner was evaluated with data chunk sizes ranging from 1,000 to 5,000 instances. The researchers observed that on average, the MR-ISVM technique outperformed the other two SVM models in accuracy, precision, and recall, and took significantly less time to train than those techniques. From the precision and recall values provided, the proposed model achieved a maximum F1 score of 0.899, while the best baseline achieved only 0.818. An additional experiment compared MR-ISVM to two SOTA techniques, where it outperformed significantly in terms of accuracy, though comparatively few details are provided on this experiment.

Rongali et al. [42]

This paper is about fraud detection in healthcare transaction, using a new specialized technique called Chaotic Iterative Gazelle-based Gated Recurrent Unit (ChaoG-GRU).

The Gated Recurrent Unit (GRU) architecture is a type of RNN frequently used in deep learning for sequence modeling tasks; in this case, healthcare transactions are represented as sequences, where each transaction occurs in a known order. The novel ChaoG algorithm is an error minimization algorithm which randomly instantiates a "herd of gazelles" (set of GRU model configurations) in the hyperparameter space, and each gazelle's "fitness" is based on classification loss. Each gazelle moves randomly and tends to travel away from its neighbors in order to avoid getting stuck in local optima. By this method, an optimal global solution for the GRU model's hyperparameters is deduced. The researchers tested their proposed model by comparing it to three other models: HEMClust [43], XGBoost, and GRU without ChaoG hyperparameter optimization. The authors claimed the dataset used was publicly available, but did not specify its source or any other properties. They used four performance metrics: F1 score, precision, accuracy, and recall. ChaoG-GRU outperformed the other models, but only marginally (0.947 F1 vs. 0.936 F1 from the next-best model).

Sowah et al. [44]

This study proposes a genetic support vector machine (GSVM) to detect fraud in health insurance claims. The GSVM operates using a SVM model, but modifies the hyperparameters iteratively using a genetic algorithm until a certain fitness is met. A genetic algorithm is an optimization technique inspired by natural selection; high-fitness (i.e., high classification performance) individuals are selected to create a future generation, while low fitness individuals are ignored or discarded. For many learners, genetic optimization will cause overfitting, but the authors argue that SVM models are resistant to overfitting. Three types of SVM learners were used, with different kernels: linear, polynomial, and radial basis function. Five subsets of data – ranging from 100 to 1000 instances – from the National Health Insurance Scheme claim dataset (obtained from hospitals in Ghana) were used to train a total of 15 versions of the proposed model, each with 10-fold cross validation. Additionally, two baseline classifiers were tested: a basic decision tree and Naïve Bayes. Accuracy, recall (a.k.a.

sensitivity), and specificity were used as performance metrics. The performance of each model varied widely over various data sizes, though the radial basis function kernel performed the best on average, and achieved the highest F1 score of 0.9948 (calculated from provided confusion matrices).

Miscellaneous supervised techniques

This category includes all other supervised classification techniques which do not fit within any of the previous categories. Note that none of these works present completely novel techniques; all papers in this section propose only domain-novel methods.

Kumaraswamy et al. [45]

In this paper, the researchers aimed to demonstrate the robustness and interpretability of a Bayesian Belief Network (BBN) model. BBNs have been used for general insurance fraud detection as early as 2003 [46]. A BBN is a directed acyclic graph where nodes represent a set of random variables, and directed edges represent statistical dependence of one node on another. The graph is manually constructed using domain knowledge and the parameters for each node are learned with training. We note that this is distinct from a knowledge graph (such as those used by the classifiers in "[Supervised learning](#)") in that graph nodes correspond to entire features (such as patient age), rather than specific patients or claims. In this study, the researchers used an anonymized Texas Medicaid outpatient prescription claims dataset with 72 million transactions. Synthetic Minority Oversampling Technique (SMOTE) was used to handle the class imbalance; however, neither the old nor new class ratios were specified. Manual feature selection was also applied for the proposed model, with twelve features chosen based on domain knowledge and prior experiments. The researchers compared three variations of their model to LR (with or without feature selection and SMOTE) and RF. The researchers found that, at best, the BBN model was comparable to the other models in performance, achieving an F1 score of 0.15 compared to the baselines' score of 0.18. However, the researchers argue that there are some key advantages to the BBN model over the other models; namely, that it is highly interpretable (as it can be visualized in graph form), and that the model is highly modifiable to incorporate any potential domain knowledge.

Levy et al. [47]

This paper is a comparison of common supervised One-Class Classification (OCC) and binary classification algorithms on credit card and Medicare fraud datasets. OCC algorithms, which train on only one class of data (usually the majority class), had been used for fraud detection before [48], but not for healthcare fraud detection. The researchers used the Medicare Part D dataset (labeled with the LEIE) and a non-healthcare fraud dataset to conduct their experiments; these datasets both are large and have severe class imbalance, particularly the Medicare data (0.07% minority-class data). Three OCC models were tested: One-Class Gaussian Mixture Model (OCGMM), One-Class Support Vector Machine (OCSVM), and One-Class Adversarial Network (OCAN). Each of these models operate by determining an expected range for non-fraudulent instances from the training data, then classifying any instances outside of this range as outliers. OCGMM and OCSVM do this using variants of Gaussian Mixture Model (GMM) and SVM models, respectively, while OCAN does this using adversarial training. The researchers trained their models with 5-fold cross-validation and evaluated model

performance using AUC and AUPRC. For non-healthcare fraud, the OCC models were only somewhat worse than the binary models, but on the Medicare data the OCC models underperformed significantly (0.729 vs. 0.969 maximum AUC). Of the OCC models, One-Class GMM had the highest performance. From these results, the researchers concluded the OCC learners often fail to recognize the minority class, and only recommend OCC when necessary; that is, when labels are available for only one class.

Non-supervised learning

In this section, we review and discuss the novel/domain-novel unsupervised and semi-supervised classifiers found in our review.

Unsupervised techniques do not train from labeled data, instead learning from patterns present in the input features alone. Many unsupervised machine learning methods do not classify data (instead acting as tools for data analysis, such as clustering); however, existing unsupervised classifiers most commonly follow an “anomaly detection” paradigm, where new instances are assigned an anomaly score based on their similarity to the training data. In the case of fraud detection, instances with anomaly scores greater than a given threshold (calculated during training) are fraudulent, and those with scores lesser than the threshold are benign. Anomaly detection methods vary primarily in their method of calculating anomaly scores; some traditional techniques include k-means or hierarchical clustering.

Because unsupervised classifiers do not use labeled data, there is less information available to learn from, theoretically limiting their performance as compared to supervised classifiers. However, unsupervised techniques hold a few key advantages over supervised techniques in this domain. For one, datasets without labels are far easier to obtain than those with. Additionally, unsupervised learners do not have to account for some of the complications which arise from using labeled data, such as class noise and (to some extent) class imbalance. Anomaly detection algorithms are also better suited for finding outlier patterns which are not represented in the training data; this is helpful for applications such as fraud detection, where new insurance fraud strategies and patterns may develop over time. A supervised model might be poorly equipped for identifying fraud done committed with a strategy not present in its training data.

Semi-supervised techniques, also called hybrid techniques, train on a mix of labeled and unlabeled data. Oftentimes, it can be difficult to use healthcare fraud datasets that are only partially labeled; unsupervised techniques cannot utilize any of the labels present, while supervised techniques cannot utilize the unlabeled instances, reducing the amount of information available in either case. Semi-supervised techniques are able to utilize both labeled and unlabeled data, typically by taking advantage of both unsupervised and supervised approaches. However, semi-supervised tasks and approaches are rarer than supervised or unsupervised ones, meaning there are often fewer baseline models to compare to.

As mentioned previously, Leevy et al. performed a review of unsupervised anomaly detection techniques in healthcare fraud detection earlier this year [4]; for the sake of avoiding redundancy, techniques discussed there will not be re-reviewed in this survey. Our review found only one paper [49] on an unsupervised classifier for healthcare fraud detection which was not covered by [4]. Additionally, our review found only one paper which presents novel/domain-novel semi-supervised classifiers within our search period.

Matloob et al. [49]

In this paper, the authors present a novel unsupervised methodology for detecting healthcare fraud using sequence mining and sequence prediction. Sequence mining and sequence prediction have been used in the medical field previously for predicting diseases [50], and the same concepts have been applied to credit card fraud detection [51], but they had not yet been utilized in the domain of healthcare fraud detection. The proposed methodology uses two modules: a Sequence Rule Engine which detects anomalous sequences, and a Prediction Based Engine which more thoroughly compares these anomalous sequences with other sequences from the training set. The researchers experimented on their method with a dataset from a local hospital in Pakistan, and the proposed technique had an accuracy of 85% on average. Baseline models and other performance metrics were not considered; the authors also did not provide any information about their dataset's class distribution, meaning the stated accuracy value may be misleading.

Bauder and Khoshgoftaar [52]

This work is a comparative study about several machine learning methods, including supervised, unsupervised, and semi-supervised learners; while most of the compared learners were not novel or domain-novel, the semi-supervised learners were both domain-novel. One technique, a “pre-training autoencoder,” first trains an unsupervised autoencoder, then uses the weights from this autoencoder in a neural network used to train the primary model (a neural network). The second technique (which the researchers simply call “hybrid”) is twofold approach utilizing a previous technique of theirs [53] named Multivariate Adaptive Regression Splines (MARS). First, a MARS model is trained on the unlabeled data, then a Bayesian probability model is trained using the prediction errors of the MARS model to predict fraud. These two models were compared to various other models, including RF, Naïve Bayes, and KNN, on a real-world, fully-labeled Medicare dataset, with either undersampling or oversampling applied to re-balance the classes. The models were evaluated using balanced accuracy, F1 score, G-measure, and Matthews Correlation Coefficient (MCC) [54]. ANOVA results showed that the pre-trained autoencoder significantly outperformed the unsupervised learners, and performed on par with many supervised learners. Meanwhile, the hybrid (MARS) model performed slightly worse than the pre-trained autoencoder, and on par with the unsupervised learners.

Discussion

In this section, we provide an in-depth analysis of the works reviewed in our survey. Of the categories established in our survey, the most common approach were supervised graph-based techniques. Of the 22 reviewed supervised methods, 8 were graph-based, 6 were meta algorithms (including 3 hyperparameter optimizers), 4 were deep learning techniques, 1 was unsupervised, 1 was semi-supervised, and 2 did not fit into any of these categories. Graph-based methods hold a plurality of 8 of 22 (~36%).

We provide a summary of our findings in Table 1, located in Appendix. We separate models by category, abbreviated to Deep Learning (DL), Graph-Based (GB), Meta-Learning (ML), Hyperparameter Optimizer (HO), Misc, or Non-Supervised (NS). We provide each work's key findings, giving the relative F1 performance of the proposed

model if this is reported or can be determined from other reported metrics. We also provide the proportion of positive instances in the dataset used by each work, and any notable methodological limitations which may compromise the reliability of the findings.

Supervised techniques

The surveyed papers demonstrated a wide range of performances. Common metrics among the works in this section include accuracy, precision, recall (a.k.a. sensitivity), F1 score, AUC, and AUPRC. There was no single metric reported by all studies; however, the most common metric was F1 score. This is presumably because F1 score is applicable to any classification model and setting, while metrics such as AUC are threshold-dependent and not applicable to certain types of models. Table 1 details the highest reported F1 score (when reported) for each proposed technique, along with the class distribution of the dataset used. Overall F1 values ranged widely, from as low as 0.15 to as high as 0.948. Statistical significance tests such as ANOVA, used to confirm that performance differences are not due to random chance, were quite rare, only conducted by two studies [37, 52]. A third study [34] provides standard deviation values for performance metrics, to a similar effect.

We also note the large range of datasets used; some datasets, like the Medicare Part B and D datasets [55], were common, while other studies used their own datasets. Class imbalance in these datasets ranged from as severe as 0.07% fraud to as much as 38.1% fraud. For datasets with high imbalance, data sampling techniques were sometimes used to rebalance the data. RUS was the most common, being used by Hancock and Khoshgoftaar [34], Shi et al. [36], and Yao et al. [37]. Other sampling techniques, were uncommon, with SMOTE (for example) only being used by Kumaraswamy et al. [45]. Other studies did not give much consideration to class imbalance, and did not even provide information about the class distribution of their dataset, such as Rongali et al. (2024) [42], Sowah et al. (2019) [44], Kumaraswamy et al. (2024) [45] and Matloob et al. (2020) [49].

When taking into account the variety of different datasets and conditions under which each classifier is evaluated, direct comparison is often difficult or impossible. The performance of a model is highly dependent on the “difficulty” of the setting in which it is evaluated, and only a few works (intentionally or unintentionally) use the same datasets and/or conditions as other works. For example, Chen et al. [20], Lu et al. [21] [22], and Hong et al. [23] test their proposed models on the same datasets (“Medical-1” and “Medical-2,” sourced from an unnamed insurance bureau) and similar techniques, and find very similar F1 scores (all within the range of 0.87–0.90). Additionally, Mayaki and Riveill [18], Mardani and Moradi [27], Zhang et al. [32], Yoo et al. [31], and Arockiam and Pushpanathan [40] also used a shared dataset (sourced from Kaggle [19]), but achieved vastly different F1 scores (as low as 0.49 [27], and as high as 0.90 [40]).

We also observe that some approaches appear to be more suited for certain datasets than for others; for example, graph-based learners yielded seemingly high F1 scores on the Medical-1 and -2 datasets [20–23], but two other graph-based learners [27, 31] significantly underperformed a hyperparameter-tuning approach [40] on the Kaggle dataset. Thus, when approaching a new task, we recommend careful consideration of what type of learner might be best for the dataset at hand. Determining this without actually

implementing each potential approach can be formulated as a machine learning problem, where the goal is to predict the optimal approach for a given dataset. This has been tried in general as early as 2002 [56], but as far as we are aware, this type of meta-analysis classification problem has not been attempted for healthcare fraud detection. Future work could look into this area.

We also note that the most common objective among reviewed supervised studies was, naturally, to demonstrate that their proposed technique outperformed some baseline models on the given task. With the graph-based techniques we found, the proposed models demonstrably improved over time, since these methods were often compared to similar and contemporary models (including other graph-based techniques also reviewed in our work). However, with the traditional and meta techniques, it is unclear if model performance is actually improving over time, since cross-comparison between recent, state-of-the-art models is rare. This reinforces the need for a comparative benchmark study which objectively compares many strong and contemporary techniques on a single dataset. This problem would additionally be mitigated by greater sharing of code and data, to facilitate model and result reproducibility. However, we find only one work [32] which provides the source code for the proposed model, and many works use private and/or proprietary datasets unavailable to other researchers. We strongly encourage future works in this domain to release model source code and data repositories, or to use publicly available datasets such as the Kaggle [19] and Medicare [55] datasets seen throughout this survey.

Although most studies aimed to create high-performance techniques, some studies did investigate other objectives. Farbmacher et al. [16] and Kumaraswamy et al. [45] focused on interpretability in their studies, and proposed models which could offer better transparency in real-world applications. Kumaraswamy et al. also addressed model robustness with respect to changing data or conditions. Future research should build on these works to investigate these objectives more thoroughly. We especially encourage further investigation into explainable models, as decision interpretability is crucial when accepting or denying healthcare insurance claims, and many methods (such as deep learning or graph-based techniques) may exhibit poor explainability without additional measures.

Finally, we remark that none of the reviewed supervised learners utilized transfer learning (a machine learning approach which applies classifiers pre-trained on other datasets). This echoes the findings of Leevy et al.'s review of unsupervised anomaly detection techniques in this domain [4], where they note that transfer learning represents a significant research gap in their surveyed works. Transfer learning has seen great success across other applications [57], and transfer learning techniques could potentially enhance model performance by taking advantage of data from other, related tasks; future research should explore supervised transfer learning methods for use in this domain.

Non-supervised techniques

We only identify one relevant unsupervised work not already discussed by Leevy et al. [4], presumably due to the short window of time between Leevy's survey and our own (approximately one year). We also find only one semi-supervised work throughout our survey period, likely due to a relative scarcity of semi-supervised tasks and strategies within the machine learning literature. Thus, there is little additional analysis possible in regards to

potential research gaps or shortcomings. Leevy et al. [4] found four primary research gaps with the current literature for unsupervised learners in this domain for which they highly recommended future work: investigation into transfer learning techniques, attention to model interpretability, work exploring online or incremental learning settings, and finally greater inter-model benchmarking and comparison. While the “semi-supervised” work we found [52] is a comparison paper, and therefore counters the latter suggestion, the two non-supervised papers discussed here otherwise do not address these concerns.

Conclusion

In this survey, we extended a previous survey by Leevy et al. [4] to review the latest methodologies for supervised, unsupervised, and semi-supervised classification techniques for healthcare insurance fraud detection. We surveyed 20 supervised, 1 unsupervised, and 1 semi-supervised classifier, for a total of 22 methods. When including the 17 papers covered by Leevy et al.’s review on unsupervised anomaly techniques for the same domain [4] (not discussed in our review), this brings the total up to 39 papers.

The surveyed studies utilize a wide range of approaches. The supervised techniques were primarily categorized between deep learning, graph-based, and meta algorithm approaches. [4] similarly finds a diverse set of unsupervised approaches, including Isolation Forest, Restricted Boltzmann Machine, and deep autoencoders.

We find that deep learning was very common overall (as was also observed in [4]). Additionally, there is little to suggest the stagnation of deep learning methods, as new techniques of this type continue to show improved performance compared to baselines. Meta-learning has also been shown to be highly performative for supervised learning. Future work could continue to look into both of these areas for healthcare fraud detection.

A significant gap in unsupervised techniques highlighted by Leevy et al. [4] is explainability and interpretability; neither of the non-supervised techniques found in our survey addressed this issue. However, we found a handful of studies on supervised methods which focused upon model explainability [16, 45]. Explainability is a critical point for the adoption of any learner in this domain, as the black-box nature of many models hinders their trustworthiness in real-world applications. Future research could focus on investigating explainability and interpretability in unsupervised and semi-supervised methods for this domain.

Other gaps in research include transfer learning and incremental learning, which none of the reviewed studies addressed. Leevy et al. [4] posit that transfer learning could be highly effective in this domain, and that incremental learning algorithms could adaptively recognize fraud as new data becomes available. Lastly, studies providing experimental comparisons of a diverse set of techniques have been performed in the recent past [58], but future work could further expand the range of methods.

The problem of healthcare insurance fraud detection still has not been fully solved by machine learning methods. Issues such as low explainability, poor benchmarking, and lack of investigation into transfer learning and incremental learning persist within supervised, unsupervised, and semi-supervised literature. By addressing these challenges, the field can move towards more reliable, transparent, and effective fraud detection systems that could better serve the healthcare insurance industry.

Appendix

See Table 1.

Table 1 Works covered in our review

Paper, category	Minority %	Key findings	Methodological limitations
[12], DL	1–9% (Sampled)	CEM improves performance of three base models. CEM+LightGBM outperformed SOTA (0.90 vs. 0.85 F1)	Data contains significant synthetic labels
[16], DL	1.5%	Proposed model shows better detection rates and simulated savings, compared to a tree-boosting model	Only one non-trivial compared baseline, no conventional metrics used
[17] ¹ , DL	50%	CNN achieves AUC of 0.70 for one fraud class, but performs poorly (≤ 0.62) for others	Significant data loss and/or overfitting from sampling, no compared baselines
[18] ^{1,2} , DL	0.05% ¹ 9.3% ^{2*}	MINN-AE inconsistent in direct comparison with simple baselines, but variants achieve overall highest AUPRC for both datasets	No contemporary compared baselines
[20] ³ , GB	1.4%, 35%	StGNN outperforms all SOTAs in node classification and anomaly detection; achieves 0.875 F1 on the latter	No notable limitations
[21] ³ , GB	1.4%, 35%	MHAMFD outperforms all SOTAs (not including StGNN [20]) in node classification and anomaly detection; 0.881 F1 on the latter	No notable limitations
[22] ³ , GB	1.4%, 35%	MBGNN outperforms all SOTAs (including StGNN [20]) in node classification and anomaly detection; 0.891 F1 on the latter	No notable limitations
[23] ³ , GB	1.4%, 35%	MHGSL outperforms all SOTAs (including StGNN [20] and MHAMFD [21]) in node classification and anomaly detection; 0.895 F1 on the latter	No notable limitations
[24], GB	0.4%	VAERM outperforms 10 SOTAs (3 of which are graph-based) in all metrics except recall. Achieves 0.928 F1	Testing datasets are artificially balanced to a 1:1 ratio (may be misleading)
[27] ² , GB	9.3%*	Best GAT model marginally outperforms graph- and decision tree-based models on testing data (0.53 F1 vs. 0.52 F1)	Claims GAT models show less overfitting, but does not provide training performance
[31] ² , GB	9.3%*	GraphSAGE outperforms LR baseline in F1 and AUC; achieves 0.49 F1	Only compares model to a single weak baseline
[32] ² , GB	38%	POCL (0.635 F1) significantly outperforms all 5 SOTAs in online tasks. Variant of POCL marginally improves (0.638 F1) at cost of time/space performance	No notable limitations
[34] ¹ , ML	50%	CatBoost outperforms SOTA significantly with geospatial information (0.890 AUC), or mildly without (0.785AUC)	Only one baseline (evaluated with different data preprocessing), significant data loss from RUS
[36] ¹ , ML	1–50% (Sampled)	Proposed NCS framework over GBM model achieves 0.782 AUC, or 0.037 F1 (results potentially inaccurate)	No baseline models, unclear/conflicting reported results
[37], ML	10–50% (Sampled)	WTBagging (max. 0.762 F1) significantly outperforms normal bagging (max. 0.756 F1) and base models (0.740 F1)	Only one non-bagging baseline
[40] ² , HO	9.3%*	MR-ISVM outperforms two other SVM models in all metrics, and two SOTAs in accuracy; achieves 0.899 F1*	Poor clarity on methodology and results of comparison to SOTAs
[42], HO	Unknown	ChaoG-GRU marginally outperforms base GRU and two SOTAs; achieves 0.948 F1	Poor clarity on dataset source/properties, and on SOTA baselines
[44], HO	Unknown	GSVM outperforms two simple baselines in terms of accuracy; best GSVM achieves 0.9948 F1*	No SOTA baselines, only accuracy presented for baselines, poor clarity on dataset properties/sampling/results

Table 1 (continued)

Paper, category	Minority %	Key findings	Methodological limitations
[45], Misc	Unknown	BBN underperforms simple baselines in F1 (0.15 vs. 0.18) and AUC (0.64 vs. 0.76), but is claimed to be more robust/interpretable (not demonstrated)	No SOTA baselines, class ratio not given, baseline result details incomplete
[47] ¹ , Misc	0.07%	All OCC models underperform all SOTA baselines in AUC (0.729 vs. 0.969) and AUPRC (0.148 vs. 0.812), but theorized to have niche when few labels available	Little to no real advancement in the literature (proposed models never show advantage over SOTA)
[49], NS	Unknown	Proposed model achieves 85% accuracy averaged between different provider specialties	No compared baselines, other metrics not given, class ratio and detailed results not given
[52], NS	20%	Semi-supervised methods (max. 0.42 F1) perform reasonably, significantly outperformed by two of four supervised SOTAs (max. 0.54 F1)	No notable limitations

Entries marked with matching superscript numbers ⁰ indicate works which use the same dataset(s). Listed minority proportions may still differ between these entries due to preprocessing differences. Asterisks * mark values which are not known precisely, but have been estimated from other information provided by the original authors

Abbreviations

ANOVA	Analysis of variance
AUC	Area under the (receiver operator characteristic) curve
AUPRC	Area under the precision-recall curve
BBN	Bayesian belief network
CEM	Credit evaluation model
CMM	Credit memories module
CNN	Convolutional neural network
DL	Deep learning
GAT	Graph attention network
GB	Graph-based
GBDT	Gradient-boosted decision tree
GMM	Gaussian mixture model
GNN	Graph neural network
GRU	Gated recurrent unit
HO	Hyperparameter optimizer
KNN	K nearest neighbors
LEIE	List of excluded individuals and entities
LR	Logistic regression
LSTM	Long short-term memory
ML	Meta-learning
MLP	Multi-layer perceptron
NS	Non-supervised
OCCAN	One-class adversarial network
OCC	One-class classification
OCCGMM	One-class Gaussian mixture model
OCSVM	One-class support vector machine
RF	Random forest
RNN	Recurrent neural network
RUS	Random undersampling
SMOTE	Synthetic minority oversampling technique
SOTA	State-of-the-art
SVM	Support vector machine

Acknowledgements

We would like to thank the reviewers in the Data Mining and Machine Learning Laboratory at Florida Atlantic University.

Author contributions

EDC performed the literature review, outlined the survey, and wrote the initial draft. PBP provided feedback and guidance throughout the writing of the manuscript, and revised the paper for publication. TMK suggested the topic to EDC and guided the research direction. All authors read and approved the final manuscript.

Funding

We acknowledge partial support by the NSF NRT-HDR (2021585). Opinions, findings, conclusions, or recommendations in this paper are the authors' and do not reflect the views of the NSF.

Data availability

No datasets were generated or analysed during the current study.

Materials availability

Not applicable.

Code availability

Not applicable.

Declarations**Ethics approval and consent to participate**

Not applicable.

Competing interests

The authors declare no competing interests.

Consent for publication

Not applicable.

Received: 26 April 2025 Accepted: 12 September 2025

Published online: 23 October 2025

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